SAT Unit 2.3.2
Exponential Functions

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Overview

1. Definition of Exponential Function

2. Graphs and Transformation

Definition and Properties

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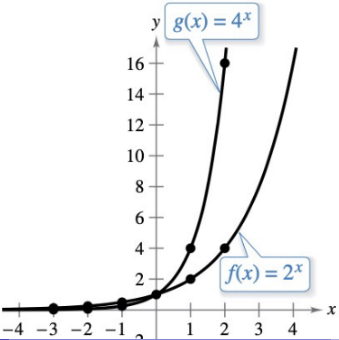
Definition of Exponential Function

Exponential Function: A function of the form:

$$f(x) = a^x$$

where:

- $a > 0$
- $a \neq 1$
- x is any real number



Key Properties of Exponential Functions

If $f(x) = a^x$, then:

- Domain: $\mathbb{R}$ (all real numbers)
- Range: $(0, \infty)$
- $f(0) = 1$ (since $a^0 = 1$)
- Continuous and smooth curve
- Horizontal asymptote at $y = 0$

Growth vs Decay:

- If $a > 1$, function exhibits **exponential growth**
- If $0 < a < 1$, function exhibits **exponential decay**

Question: Identifying k in an Exponential Function

For the exponential function f, the value of $f(1)$ is k, where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

Options:

A. $f(x) = 50(2)^{x+1}$
B. $f(x) = 80(2)^{x}$
C. $f(x) = 128(2)^{x-1}$
D. $f(x) = 205(2)^{x-2}$

Answer and Explanation

We are told $f(1) = k$. Let's plug $x = 1$ into each option and see which form gives $f(1) = k$ as the coefficient or base.

- **A.** $f(1) = 50(2)^2 = 50 \cdot 4 = 200$
- **B.** $f(1) = 80(2)^1 = 160$
- **C.** $f(1) = 128(2)^0 = 128 \cdot 1 = 128$
- **D.** $f(1) = 205(2)^{-1} = \frac{205}{2} = 102.5$

Only in option **C**, we get $f(1) = 128$, and the coefficient **is** k directly.

Correct Answer: C

Question: Exponential Growth Relationship

Two variables, x and y, are related such that for each increase of 1 in the value of x, the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

Options:

A. $y = 4(x)^{200}$

B. $y = 4(200)^x$

C. $y = 200(2)^x$

D. $y = 200(4)^x$

Answer and Explanation

This is an exponential relationship where y increases by a factor of 4 for every 1 increase in x. The general exponential form is:

$$y = a \cdot b^x$$

Given:

- Initial value: $y = 200$ when $x = 0$
- Growth factor: $b = 4$

So the equation becomes:

$$y = 200(4)^x$$

Correct Answer: D

Question: Finding Constant in Exponential Function

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = f(x)$ has:

- an x-intercept at $(2, 0)$
- a y-intercept at $(0, -323)$

What is the value of b?

Answer and Explanation

We are given two points:

- $f(2) = 0 \Rightarrow a^2 + b = 0$
- $f(0) = -323 \Rightarrow a^0 + b = -323$

From the second equation:

$$1 + b = -323 \Rightarrow b = -324$$

Substitute $b = -324$ into the first equation:

$$a^2 - 324 = 0 \Rightarrow a^2 = 324 \Rightarrow a = 18$$

Final Answer: $b = -324$

Graphs and Transformation

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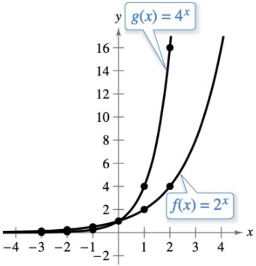
Graphs of Exponential Functions

Graphs of $y = a^x$

In the same coordinate plane, sketch the graph of each function.

1. $f(x) = 2^x$
2. $g(x) = 4^x$

Begin by constructing a table of values.

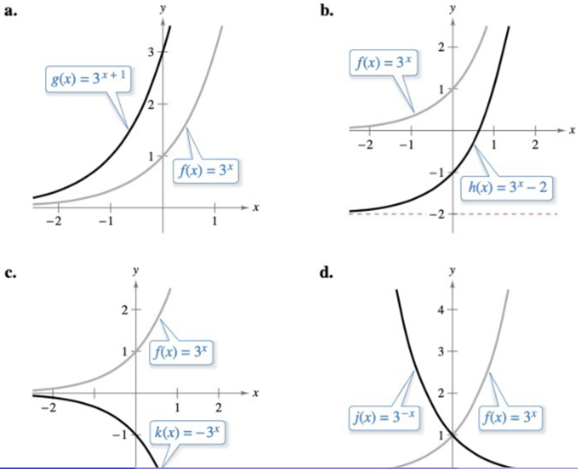


Observation:

- Exponential functions increase/decrease rapidly
- All graphs pass through the point (0, 1)

Transformations of the Exponential Function

Parent function: $f(x) = 3^x$ (gray curve in each picture below)



Transformations of the Exponential Function — Quick Rules

Start with $f(x) = 3^x$.

ponent $f(x+1) = 3^{x+1}$
Replace x with $x+1 \Rightarrow$ graph moves **left** 1 unit.

umber $f(x) - 2 = 3^x - 2$
Subtract 2 from the output $\Rightarrow$ graph moves **down** 2 units.

front $-f(x) = -3^x$
Multiply outputs by $-1 \Rightarrow$ **reflection across the x-axis**.

input $f(-x) = 3^{-x}$
Replace x with $-x \Rightarrow$ **reflection across the y-axis**.

Memory trick: changes **inside** the function move the graph **left/right**; changes **outside** move it **up/down**.

Question: X-Intercepts of an Exponential Function

Given the equation:

$$y = 9\left(\frac{a}{6}\right)^{x+c} - b,$$

how many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 6$ and $b > c$?

Options:

A. Zero

B. One

C. Two

D. Three

Answer and Explanation

We are given the exponential function:

$$y = 9\left(\frac{a}{6}\right)^{x+c} - b$$

- Since $a > 6$, $\frac{a}{6} > 1$, meaning the function is **increasing**.
- The expression $9\left(\frac{a}{6}\right)^{x+c}$ is always positive.
- As $x \to -\infty$, $\left(\frac{a}{6}\right)^{x+c} \to 0$, and $y \to -b$, so the horizontal asymptote is $y = -b$.
- As x increases, y increases beyond $-b$.
- Since $y = 0$ when $9\left(\frac{a}{6}\right)^{x+c} = b$, and this equation has exactly one solution for x, the graph crosses the x-axis exactly once.

Correct Answer: B. One

Applications of Exponential Functions

Exponential functions are used to model real-world situations like:

- **Population Growth:** $P(t) = P_0 \cdot a^t$
- **Compound Interest:** $A = P(1 + r)^t$
- **Radioactive Decay:** $M(t) = M_0 \cdot a^{-t}$
- **Bacterial Growth**

Example: A population doubles every 3 years:

$$P(t) = 100 \cdot 2^{t/3}$$

Exponential Function with Time Units — Question

The given function $p(t) = 90{,}000(1.06)^t$ models the population of Lowell t years after a census.

Which of the following functions best models the population of Lowell m months after the census?

A. $r(m) = \dfrac{90{,}000}{12}(1.06)^m$

B. $r(m) = 90{,}000\left(\dfrac{1.06}{12}\right)^m$

C. $r(m) = 90{,}000\left(\dfrac{1.06}{12}\right)^{\frac{m}{12}}$

D. $r(m) = 90{,}000(1.06)^{\frac{m}{12}}$

Exponential Function in Terms of Different Units

The original exponential model is:

$$p(t) = 90{,}000(1.06)^t$$

where t is measured in **years**.

To express the model in terms of **months**, we must use:

$$t = \frac{m}{12}$$

Substitute into the original function:

$$r(m) = 90{,}000(1.06)^{\frac{m}{12}}$$

This gives us the population model in terms of months.

Correct Answer: D

$$\boxed{r(m) = 90{,}000(1.06)^{\frac{m}{12}}}$$

Exponential Modeling — Question

The function P models the population, in thousands, of a certain city t years after 2005:

$$P(t) = 290(1.04)^{\left(\frac{4}{6}\right)t}$$

According to the model, the population is predicted to increase by $n\%$ every 18 months.

What is the value of n?

Choices:

A. 0.38

B. 1.04

C. 4

D. 6

Exponential Modeling — Solution

Given:

$$P(t) = 290(1.04)^{\left(\frac{4}{6}\right)t}$$

Step 1: Convert 18 months to years

$$18 \text{ months} = \frac{18}{12} = 1.5 \text{ years}$$

Step 2: Plug in $t = 1.5$ to evaluate the growth over 18 months

$$P(1.5) = 290(1.04)^{\left(\frac{4}{6}\right)(1.5)} = 290(1.04)^1$$

Conclusion: The population grows by a factor of 1.04 every 18 months, or **4%**.

Answer: $\boxed{\text{C. } n = 4}$

Question

The function

$$f(t) = 20\left(1 + \frac{15}{100}\right)^t$$

models the temperature, in degrees Celsius, of a metal alloy used in an experiment, where t is the number of seconds after the experiment began. Which of the following is the best interpretation of the number 15 in this context?

A) The temperature, in degrees Celsius, of the metal alloy at the beginning of the experiment

B) The increase in the temperature, in degrees Celsius, of the metal alloy every 100 seconds during the experiment

C) The percent by which the temperature, in degrees Celsius, of the metal alloy decreased from each second to the next during the experiment

D) The percent by which the temperature, in degrees Celsius, of the metal alloy increased from each second to the next during the experiment

Answer

The function is:

$$f(t) = 20\left(1 + \frac{15}{100}\right)^t = 20(1.15)^t$$

This is an exponential growth function of the form:

$$f(t) = a(1 + r)^t$$

Where $a = 20$ is the initial value, and $r = 0.15$ or 15% is the percent increase per second.
Thus, the number 15 represents the **percent increase per second** in temperature.
Correct answer: D

Question

The population of trees in a forest has been decreasing by 6 percent every 4 years. The population at the beginning of 2015 was estimated to be 14,000.
If P represents the population of trees in the forest t years after 2015, which of the following equations best defines P?

A) $P = 14{,}000(0.06)^{\frac{t}{4}}$
B) $P = 14{,}000 + 0.94(4t)$
C) $P = 14{,}000(0.94)^{4t}$
D) $P = 14{,}000(0.94)^{\frac{t}{4}}$

Answer

We are told the population decreases by 6% every 4 years.
This is exponential decay, so the decay factor is:

$$1 - 0.06 = 0.94$$

Every 4 years, the population is multiplied by 0.94. Since t is in years, and 0.94 applies every 4 years, the exponent must be $\frac{t}{4}$.
So the equation is:

$$P = 14{,}000(0.94)^{\frac{t}{4}}$$

Correct answer: D

Question

The amount of data, in gigabytes, stored in a database increases by 2% every 15 hours. If 16 gigabytes worth of data are currently stored in the database, which of the following functions g gives the amount of data, in gigabytes, that will be stored in the database t days from now?

A) $g(t) = 16(2)^{15t}$

B) $g(t) = 16(1.02)^{\frac{t}{15}}$

C) $g(t) = 16(1.02)^{\frac{8t}{5}}$

D) $g(t) = 16(1.02)^{\frac{5t}{8}}$

Answer

The data increase by 2% every 15 hours. In t days, there are $24t$ hours, so the number of 15-hour periods is:

$$\frac{24t}{15} = \frac{8t}{5}$$

Therefore:

$$g(t) = 16(1.02)^{\frac{8t}{5}}$$

Correct answer: C

Question

The exponential function g is defined by the given equation:

$$g(x) = 256(0.61)^{\frac{x}{30}}$$

The equation can be rewritten as:

$$g(x) = 256\left(1 - \frac{k}{100}\right)^{\frac{x}{10}},$$

where k is a constant.
What is the value of k, to the nearest whole number?

A) 15
B) 39
C) 61
D) 77

Answer

We are given:

$$g(x) = 256(0.61)^{\frac{x}{30}} = 256\left(1 - \frac{k}{100}\right)^{\frac{x}{10}}$$

Let:

$$r = 1 - \frac{k}{100} \Rightarrow \left(1 - \frac{k}{100}\right)^{\frac{x}{10}} = (0.61)^{\frac{x}{30}}$$

So:

$$r^{\frac{x}{10}} = (0.61)^{\frac{x}{30}} \Rightarrow r^{1/10} = (0.61)^{1/30}$$

Raise both sides to the 10th power:

$$r = \left((0.61)^{1/30}\right)^{10} = (0.61)^{\frac{1}{3}}$$

Now compute $(0.61)^{1/3} \approx 0.850$ (cube root of 0.61)
So:

$$1 - \frac{k}{100} \approx 0.85 \Rightarrow \frac{k}{100} \approx 0.15 \Rightarrow k \approx 15$$

Correct answer: A

Question

The function

$$W(t) = 830(1.09)^{\frac{6}{8}t}$$

gives the value, in dollars, of a certain investment after t years.
If the value of the investment increases by 9% every n months, what is the value of n?

Answer

We are given:

$$W(t) = 830(1.09)^{\frac{6}{8}t}$$

The investment increases by 9% each time the exponent increases by 1. Let m be the number of months. Since $t = \frac{m}{12}$:

$$\frac{6}{8}t = \frac{6}{8} \cdot \frac{m}{12} = \frac{m}{16}$$

So the value grows by 9% every 16 months.

$$\boxed{16}$$

Final Answer: $n = 16$

Question

To examine how a certain virus spreads, scientists introduced the virus to cells in a test tube and found that the number of infected cells in the test tube grew exponentially over time.
The given function C models the number of infected cells in the test tube t days after the virus was introduced:

$$C(t) = 100(2)^{\frac{t}{5}}$$

Which is the best interpretation of $(2)^{\frac{t}{5}}$ in this context?

A) The predicted number of infected cells in the test tube doubled every 5 days.
B) The predicted number of infected cells in the test tube grew by a factor of 5 every two days.
C) The predicted number of infected cells in the test tube doubled every day.
D) The predicted number of infected cells in the test tube grew by a factor of 5 every day.

Answer

The function is:

$$C(t) = 100(2)^{\frac{t}{5}}$$

This is an exponential growth function with base 2 and exponent $\frac{t}{5}$, which means the quantity doubles every 5 days. That's because:

$$C(5) = 100(2)^1 = 200 \Rightarrow \text{doubling in 5 days}$$

Correct answer: A

Question

The function f is defined for all $x \geq 0$.
Which of the following equivalent forms of the function f displays, as a constant or coefficient, the minimum value of f, where $x \geq 0$?

A) $f(x) = 50(2.5)(1.25)^{x-2}$

B) $f(x) = 125(0.8)(1.25)^{x-1}$

C) $f(x) = 64(1.25)^{x+1}$

D) $f(x) = 80(0.64)(1.25)^{x+2}$

Answer

We are asked to identify the expression that makes the **minimum value of** $f(x)$ appear as a constant or coefficient, given that $x \geq 0$.
For an exponential function in the form:

$$f(x) = A \cdot B^{x+c}$$

The minimum occurs when the exponent is minimized (i.e., $x = 0$).
Let's test $x = 0$ in all options:
A) $f(0) = 50(2.5)(1.25)^{-2} = 125 \cdot (1/1.5625) \approx 80$ **B)** $f(0) = 125(0.8)(1.25)^{-1} = 100 \cdot (1/1.25) = 80$ **C)** $f(0) = 64(1.25)^{1} = 80$ **D)** $f(0) = 80(0.64)(1.25)^{2} = 80(0.64)(1.5625) = 80$
All give the same result for $x = 0$: $f(0) = 80$. But the key is: which one displays the **minimum value directly as a constant or coefficient**?
Only **option D** has:

$$f(x) = \boxed{80}(0.64)(1.25)^{x+2}$$

and no exponent adjustment on the 80 — it's clearly the base value when $x = 0$.
Correct answer: D

Thank You!

We appreciate your time and focus.

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